Current Affairs 26-10-2026

Credit - Drishti Ias

Media Regulation and the Need for National Media Council

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For Prelims: Public Interest Litigation , OSHWC Code, 2020 , Press Council of India (PCI) , News Broadcasting & Digital Standards Authority (NBDSA) , Central Board of Film Certification (CBFC)

For Mains: Journalistic ethics and professional integrity, Freedom of speech, Constitutional safeguards, Media governance reforms, Media accountability and civil liberties.

Source: TH

Why in News?

The debate over establishing a **National Media Council (NMC)** has gained fresh momentum following the crackdown on freelance journalists and content creators during the **October 2026 protests in New Delhi.**

- The debate has further intensified as the **Supreme Court,** while hearing a **Public Interest Litigation (PIL)** concerning the protection of freelance journalists, raised questions about the definition of a journalist in the digital age and highlighted the **need for a national protocol and a unified media regulatory framework.**
- In 2024, the **Press Club of India** drafted the **Media Transparency and Accountability Bill,** proposing the establishment of an NMC, among other reforms.

Summary

- Rising calls for a National Media Council (NMC) reflect the need for a unified regulatory framework to protect journalists, improve media accountability and address challenges posed by digital media.
- Growing concerns over press freedom and misinformation highlight the need for independent media regulation that balances accountability with constitutional protection under Article 19(1)(a).

What is the Current Framework for Media

Regulation and Recognition?

- **Defining a Journalist:** Under the **Occupational Safety, Health and Working Conditions (OSHWC) Code, 2020** , a working journalist is someone whose principal avocation is journalism in print, electronic, or digital media, excluding those in managerial roles.
- **Government Accreditation:** The Central and State governments grant accreditation via bodies like the **Press Information Bureau (PIB)** and the **Directorate of Information and Publicity (DIP)** .
  - However, this is governed by **strict quotas based on a newspaper's circulation, a news agency's subscriber base, or a digital portal's unique visitors.**
  - Consequently, many genuine independent and freelance working journalists fail to secure accreditation.
- **Institutional Regulation:**
  - **Print Media:** Regulated by the **Press Council of India (PCI)** , a statutory, quasi-judicial autonomous body established under the Press Council Act, 1978.
    - It acts as a moral watchdog but has **no punitive powers** to enforce its guidelines.
  - **Electronic/Television Media:** Primarily self-regulated through bodies like the **News Broadcasting & Digital Standards Authority (NBDSA)** , set up by the **News Broadcasters & Digital Association (NBDA),** independent of statutory government control.
    - Television Broadcasting Regulation governed by the **Cable Television Networks (Regulation) Act, 1995,** and the Programme and Advertising Codes under the 1994 Rules. The 2021 amendments introduced a **three-tier grievance redressal mechanism** involving broadcasters, self-regulatory bodies and Central Government oversight.
  - **Digital Media and OTT:** Governed by the **Information Technology (Intermediary Guidelines and Digital Media Ethics Code) Rules, 2021** , which provide a **three-tier grievance** redressal mechanism but face criticism for excessive executive interference.
  - **Films/Cinema:** Regulated by the **Central Board of Film Certification (CBFC)** under the Cinematograph Act, 1952.
  - **Online Content Regulation:** Governed by the **Information Technology Act, 2000.** Section 69A empowers the Central Government to block online content on specified grounds, while **Section 79(3)(b) concerns intermediaries' safe-harbour protection** when they fail to act against unlawful content after legally valid notice.
    - The **Sahyog portal** facilitates communication between authorised government agencies and online platforms.

What is the Need for the National Media Council (NMC)?

- **The Crisis of Defining a "Journalist":** The proliferation of smartphones has blurred the line between professional journalists and content creators.
  - The Solicitor General noted that many individuals claim to be journalists merely to create **social media reels for monetary gain** by sensationalising events.
  - An NMC could establish clear, professional parameters to distinguish genuine journalists from bad actors.
- **Need for a Professional Licensing Body:** Currently, professions like law and accounting are regulated by statutory bodies (the **Bar Council of India** and **Institute of Chartered Accountants of**

India (ICAI) respectively).

- A MNC could act similarly, **mandating a degree or diploma in journalism to issue professional licenses and recognize** all working journalists, freelancers, and camerapersons uniformly, bypassing the government quota system.

- **Tackling Media Convergence:** A single corporate entity today operates print, television, and digital media simultaneously.
  - Regulating them through separate **silos (PCI, NBDSA, IT Rules)** creates administrative loopholes.
  - An NMC would serve as a **single, unified statutory authority reflecting the modern converged media landscape.**
- **Empowering the Watchdog:** The existing PCI is widely considered a "toothless tiger" because its jurisdiction excludes TV and digital media, and it lacks the authority to impose fines, suspend licenses, or enforce primetime apologies for severe ethical breaches (such as media trials or coordinated fake news).
- **Failure of Self-Regulation in TV Media:** Electronic media's self-regulatory models have largely failed to **curb sensationalism,** toxic debates, prioritizing TRPs **(Television Rating Points)** over journalistic ethics.
- **Fake News and Misinformation:** The rapid proliferation of digital platforms and **AI-driven deepfakes** has exacerbated the problem of coordinated misinformation, necessitating a comprehensive statutory mechanism that covers all mediums.
- **Cross-Media Ownership:** There is a growing threat of **corporate and political monopolies over media houses,** affecting editorial diversity, while engagement-driven digital algorithms can amplify sensationalism, misinformation and polarising content.
  - A modern regulatory framework could promote cross-media ownership transparency, disclosure of conflicts of interest, algorithmic accountability and media pluralism.

What are the Challenges in Establishing an Independent National Medical Commission (NMC)?

- **Appointments & Executive Overreach:** Government control over member appointments risks **turning the NMC into a censorship tool.**
  - Ensuring true independence requires a multi-partisan or judicial collegium, which governments are often reluctant to establish.
- **Financial Autonomy:** Relying on the Ministry of Information and Broadcasting (MIB) for funding compromises operational independence, yet **alternative models (like a corporate media cess) face heavy industry resistance.**
- **Government Advertising Leverage:** State and Union governments frequently use the allocation or withdrawal of government advertising funds **(via the Directorate of Advertising and Visual Publicity - DAVP)** to engineer editorial compliance.
  - Even if an independent NMC is established, it will struggle to protect media autonomy if the government retains this financial leverage.
  - As the Supreme Court ruled in the landmark ***Bennett Coleman & Co. vs. Union of India (1972)* case** , policies causing the financial strangulation or limiting the economic viability of the press directly violate **Article 19(1)(a).** The Court termed the **"Ark of the Covenant of Democracy"** from such covert financial censorship.
- **Constitutional Hurdles:** Granting the **NMC punitive powers (fines/suspensions) or mandating**

professional licenses risks gatekeeping.

  - This could stifle **investigative journalism and conflict with the fundamental right to free speech under Article 19(1)(a).**

- **Corporate Resistance:** Large media conglomerates strongly prefer current self-regulatory models (like NBDSA).
  - They are likely to lobby against an NMC that threatens their commercial interests, TRP-driven models, and cross-media monopolies.
- **Digital Complexity:** The modern **"creator economy"** blurs the line between a professional journalist and a social media content creator. Furthermore, borderless digital platforms create severe jurisdictional hurdles for enforcing NMC rulings.

What Measures are Needed for Effective Journalism?

- **Establishing a National Police Protocol:** As argued by legal experts in the Supreme Court, there is an immediate need for a **standardized national protocol** directing law enforcement not to obstruct journalists who carry valid identity cards from their employers, regardless of PIB/DIP accreditation.
- **Independent Co-Regulation Model:** The Parliamentary Standing Committee on Communications and Information Technology, in its **27 th Report (2021) on 'Ethical Standards in Media Coverage'** , highlighted the **need to balance regulation with free speech.**
  - Instead of direct government control, the NMC should operate on a **"co-regulation" model (a mix of statutory backing and self-regulation).** Its selection committee must be insulated from the executive, ideally comprising the Chief Justice of India, the Leader of the Opposition, and eminent media veterans.
- **Regulating Cross-Media Ownership:** As strongly recommended by the Telecom Regulatory Authority of India (TRAI) in its **2014 report on "Issues relating to Media Ownership"** , strict regulations must be enforced to restrict cross-media monopolies.
- **Inclusive Standard Setting:** While the NMC can issue licenses or press cards to formally recognize freelancers and protect them during protests, it must explicitly ensure that holding such a license is a *privilege* for access, not a *mandatory requirement* to publish news, thereby safeguarding the constitutional right to free speech.
- **Anti-SLAPP Legislation:** Enact laws to protect journalists from **SLAPP (Strategic Lawsuits Against Public Participation)** suits, which are frivolous defamation cases filed by powerful entities to drain journalists financially and stifle investigative reporting.
- **Curbing Fake News:** The 2025 Parliamentary Standing Committee on Communications and IT recommended fact-checking mechanisms, internal media ombudsmen, stricter penalties and a clear legal definition of fake news, while safeguarding freedom of speech.
- **Independent News Ombudsman:** Encourage media organisations to appoint independent ombudsmen to address complaints about accuracy, fairness, bias and ethical misconduct, promoting self-regulation, transparency and public trust.
- **Adopting Global Best Practices:**
  - **Australia’s News Media Bargaining Code:** Mandates Big Tech to pay news publishers for content, ensuring the financial sustainability of independent media.
  - **EU’s Digital Services Act (DSA):** Holds large online platforms accountable for algorithmic transparency and curbing fake news, hate speech, and election manipulation.

Conclusion

The ultimate goal of these measures is to strike a delicate balance between **Article 19(1)(a)** (Freedom of Speech and Expression) and **Article 19(2)** (Reasonable Restrictions). By safeguarding reporters from intimidation and insulating newsrooms from corporate-political nexus, India can ensure **that its media remains a fearless watchdog of democracy.**

Drishti Mains Practice Question

Critically examine the limitations of existing media regulatory institutions in addressing the challenges of digital media convergence.

Frequently Asked Questions (FAQs)

1. What is the proposed National Media Council (NMC)?
A proposed unified media regulatory body intended to oversee print, television and digital media while strengthening accountability and protecting journalistic freedom.

2. What is the Press Council of India (PCI)?
A statutory, quasi-judicial autonomous body established under the Press Council Act, 1978, to preserve press freedom and maintain standards in print media.

3. Which constitutional provisions govern press freedom in India?
Article 19(1)(a) guarantees freedom of speech and expression, while Article 19(2) permits reasonable restrictions on specified grounds.

4. What are SLAPP suits?
Strategic Lawsuits Against Public Participation are legal actions used to intimidate or financially burden journalists and discourage public-interest reporting.

5. What measures can strengthen media accountability in India?
Establishing an independent co-regulatory NMC, adopting a national police protocol, regulating cross-media ownership, enacting anti-SLAPP safeguards and strengthening fact-checking mechanisms.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/FVElMECP4p4**]

UPSC Civil Services Examination Previous Year Question (PYQ)

Mains

Q. What do you understand by the concept of “freedom of speech and expression”? Does it cover hate speech also? Why do the films in India stand on a slightly different plane from other forms of expression? Discuss. **(2014)**

Biomarkers (Chemical Signatures) and Early Disease Detection

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Source: TH

Why in News?

Scientists have emphasised the growing potential of **chemical signatures** (commonly known as **biomarkers**) in revolutionising early disease diagnosis, particularly for neurological conditions like **Alzheimer 's and Parkinson's .**

- However, despite their promise, integrating these diagnostic tools into routine clinical practice remains hindered by significant hurdles related to accuracy, cost, and accessibility.

What are Chemical Signatures (Biomarkers)?

- **About:** Chemical Signature is a measurable **biological indicator** found in bodily fluids (like blood or cerebrospinal fluid) consisting of a unique pattern of specific proteins, molecules, elements, isotopes, or by-products.
- **Function:** It objectively signifies what is happening inside the body, indicating a normal biological process, a pathogenic disease state, or a response to medical treatments.
- **Working of Chemical Signatures:** When a disease develops, it triggers changes at the **molecular level** long before physical symptoms appear.
  - By measuring the presence, absence, or altered behaviour of specific proteins, doctors can identify underlying physiological shifts.
- **Detection Instruments:** Advanced analytical tools like **spectrometers** (which measure molecules by mass or light absorption) and **chromatographs** (which separate mixture components) are used to detect these minute chemical patterns.
- **State of Biomarkers in India:**
  - **GenomeIndia Project:** Launched in January 2020 and funded by the Department of Biotechnology (DBT), the project has completed the **whole-genome sequencing of over 10,000 individuals** representing diverse Indian populations.
    - As one of the **world's largest genetic databases** , **Genome India Project** serves as a critical foundation for discovering population-specific biomarkers and genetic risk factors tailored to South Asians.
  - **BHARAT Study:** The Indian Institute of Science (IISc), under its Longevity India Programme, has launched the **BHARAT (Biomarkers of Healthy Aging, Resilience, Adversity, and Transitions) Study** to establish a “Bharat Baseline” tailored to the Indian population, reducing reliance on Western-derived biomarkers.
    - Since ageing is influenced by **molecular, cellular, environmental and socioeconomic factors, chronological age** may not accurately reflect biological age. The study aims to develop biomarkers that better capture ageing **patterns among Indians.**

Significance of Biomarkers in Healthcare

- **Early and Pre-symptomatic Detection:** Biomarkers allow physicians to transition from late-stage symptomatic management to early molecular triage, detecting conditions years before cognitive or physical decline begins.
- **Non-Invasive Diagnostics:** Blood-based biomarker assays offer a cheaper, safer, and less painful alternative to invasive procedures like lumbar punctures (spinal taps) or expensive nuclear **PET (Positron Emission Tomography) scans** .
- **Targeted Therapies:** By differentiating between diseases with overlapping symptoms (e.g., differentiating types of dementia), biomarkers ensure patients receive the correct targeted therapies rather than broad-spectrum symptom suppressants.
  - India’s precision medicine market is projected to grow at a CAGR of 16%, exceeding USD 5 billion by 2030. Biomarkers can accelerate this growth by personalised treatments and more targeted therapies.
- **Tracking Disease Progression:** Serial blood tests allow doctors to monitor how well a patient is responding to experimental drugs or ongoing treatments dynamically over time.

Common Biomarkers and Linked Diseases

Biomarker	Associated Disease	Clinical Indication
Tau (p-tau217) & Amyloid-beta	Alzheimer’s Disease	Tracks the buildup of abnormal plaques and neurofibrillary tangles in the brain.
Alpha-synuclein	Parkinson’s Disease	Detects misfolded protein clumps linked to the loss of nerve cells that control movement.
Neurofilament Light (NfL)	Multiple Neurological Disorders	Acts as a non-specific marker indicating nerve cell damage and neuroaxonal injury.
Neuronal Surface Antibodies	Autoimmune Encephalitis	Indicates the immune system mistakenly attacking brain proteins, aiding in reversible neuro-inflammatory treatment.

Major Global Genome Sequencing Projects

Project	Country/Region	Objective
All of Us Research Program	**USA**	Advance personalized medicine using genomic and health data.
Human Microbiome Project	**USA**	Study human-associated microbes and their impact on health.
100,000 Genomes Project	**UK**	Diagnose rare diseases and cancers through whole-genome sequencing.
1+ Million Genomes (1+MG)	**European Union**	Enable secure genomic data sharing for research.
Emirati Genome Program	**UAE**	Improve preventive and personalised healthcare.

What are the Challenges in Testing and Diagnosis through Biomarkers?

- **Accuracy and Specificity Issues:** Many biomarkers lack absolute specificity. For instance, the **Neurofilament Light (NfL) protein** indicates nerve damage but cannot pinpoint the exact disease causing it.
  - Furthermore, factors like age, kidney function, and other concurrent medications can alter biomarker levels, leading to a risk of false-positive or false-negative results.
- **High Costs and Lack of Infrastructure:** Specialised tests requiring advanced **immunoassays are expensive and largely confined to major metropolitan centres.**
  - District and rural hospitals often lack the automated platforms required for these tests, resulting in severe diagnostic deficits in rural India.
- **Lack of Uniform Standards:** There is a lack of harmonisation across testing laboratories.
  - Variations in laboratory methods and equipment can affect the consistency and interpretation of biomarker test results.
- **Ethical and Socio-economic Concerns:** Identifying preclinical diseases in asymptomatic individuals creates "patients-in-waiting," triggering severe psychological anxiety.
  - Additionally, a positive biomarker test on medical records could expose individuals to socio-economic discrimination, such as denial of life insurance or workplace bias.

Way Forward

- **Promoting Indigenous R&D:** The government and public sectors must support domestic biotechnology firms to engineer affordable, locally manufactured diagnostic reagents. This will reduce

dependency on costly imported testing platforms and democratise access.

- **Standardisation of Protocols:** The **Indian Council of Medical Research (ICMR)** and the **Central Drugs Standard Control Organisation (CDSCO)** should establish unified clinical guidelines and standard operating procedures (SOPs) for biomarker testing to ensure uniform laboratory interpretation.
- **Ethical Safeguards:** Clear legal frameworks must be established regarding genetic and biomarker data privacy to prevent insurance and employment discrimination against asymptomatic 'at-risk' individuals.
- **Capacity Building:** Expanding diagnostic infrastructure at the district hospital level and training medical professionals in biomarker interpretation is critical for integrating these tools into mainstream public health planning.

Frequently Asked Questions (FAQs)

1. What are chemical biomarkers?
Measurable biological indicators used for disease detection, diagnosis and monitoring.

2. Which biomarkers are linked to Alzheimer’s and Parkinson’s?
Tau and Amyloid-beta are linked to Alzheimer’s, while Alpha-synuclein is linked to Parkinson’s.

3. What are the major challenges in biomarker testing?
High costs, accuracy issues, inadequate infrastructure and data privacy risks.

4. What are the roles of ICMR and CDSCO?
ICMR supports biomedical research, while CDSCO regulates drugs and medical devices.

5. How can India improve biomarker-based diagnosis?
Through indigenous R&D, standardised testing, affordable diagnostics and stronger rural healthcare infrastructure.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/9urfvJDk7YM**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Which one of the following statements best describes the role of B cells and T cells in the human body? (2022)

(a) They protect the environmental allergens. body

(b) They alleviate the body's pain and inflammation.

(c) They act as immunosuppressants in the body.

(d) They protect the body from diseases caused by pathogens.

Ans: (d)

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future? (2017)

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.

Select the correct answer using the code given below:

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Ans: (d)

Mains

Q. Why is there so much activity in the field of biotechnology in our country? How has this activity benefitted the field of biopharma? **(2018)**

India's Wetland Conservation Journey

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Source: PIB

Why in News?

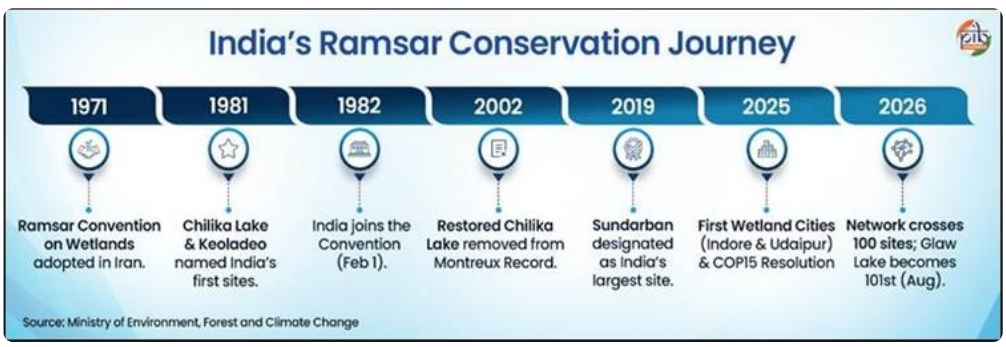
India's network of **Ramsar Sites has expanded to 103 wetlands in October 2026** , with **Sukhna Wetland in Chandigarh** and **Sauj Jheel in Uttar Pradesh** as the latest additions, strengthening the country's efforts towards biodiversity conservation, water security and climate resilience.

What are the Key Facts About Ramsar Sites in India?

- **About:** Ramsar Sites are wetlands designated as **Wetlands of International Importance under the Ramsar Convention** because of their ecological, botanical, zoological, limnological or hydrological significance.
- **Ramsar Convention:** An **intergovernmental treaty adopted in 1971 at Ramsar,** Iran, which **entered into force in 1975** to promote the conservation and wise use of wetlands.
  - It has **172 Contracting Parties and over 2,536 Ramsar Sites** globally as of October 2026.
- **India's Membership:** India became a Contracting Party to the Ramsar Convention **in 1982.**

India's Diverse Wetland Ecosystems

- India’s Ramsar Sites include **high-altitude lakes, river floodplains, mangroves , coral reefs , village tanks and urban marshes** , reflecting the country’s diverse wetland ecosystems.
- **Major Achievements:**
  - India's Ramsar network increased from **26 sites in 2014 to 103 in October 2026,** covering over 13.87 lakh hectares across 28 States and Union Territories.
  - India ranks **first in Asia and third globally in terms of the number of Ramsar Sites.**
  - **Tamil Nadu has the highest number of Ramsar Sites (20), followed by Uttar Pradesh (14).**
- **First Ramsar Sites: Chilika Lake (Odisha)** and **Keoladeo National Park (Rajasthan)** were India's first Ramsar Sites, designated in 1981.
- **Largest Ramsar Site: Sundarban Wetland (West Bengal),** covering approximately 4.23 lakh hectares. It holds over 60 per cent of India’s mangrove area and a tiger population with unique aquatic hunting skills.
- **Chilika Lake Restoration: Chilika Lake was removed from the Montreux Record in 2002** , and the **Chilika Development Authority** later received the **Ramsar Wetland Conservation Award** .
- **Wetland City Accreditation:** Indore (Madhya Pradesh) and Udaipur (Rajasthan) became **India's first Ramsar-accredited Wetland Cities in January 2025** , recognising urban wetland conservation.



What are the Key Government Initiatives for Wetland Conservation?

- **Wetlands (Conservation and Management) Rules, 2017 :** Notified under the **Environment (Protection) Act, 1986** , these rules establish State/UT Wetlands Authorities and prohibit activities such as encroachment, solid-waste dumping and discharge of untreated effluents into notified wetlands.
- **National Plan for Conservation of Aquatic Ecosystems (NPCA)** : A **centrally sponsored scheme** integrating the conservation of lakes and wetlands. The **revised NPCA Guidelines, 2024** promote scientific, integrated and sustainable wetland management.
- **Amrit Dharohar :** Promotes the conservation values of Ramsar Sites through species and habitat conservation, nature tourism, livelihood development and wetland carbon assessment.
  - Inventories of **flora and fauna have been undertaken for 75 Ramsar Sites,** while carbon stock assessments have been conducted at 39 sites.
- **Mission Sahbhagita:** Promotes community-led wetland conservation through public participation and ecological awareness.
  - Under the **Save Wetlands Campaign** , more than 20 lakh volunteers have helped map over 1.7 lakh wetlands.
- **Mangrove Initiative for Shoreline Habitats and Tangible Incomes (MISHTI) :** Targets the restoration of 540 sq. km of mangroves across nine States and four UTs, strengthening coastal resilience and livelihoods.
- **Wetlands of India Portal:** Launched in **2021, it provides a digital repository of wetland information, conservation initiatives, research and training resources** .

What are the Key Features of the Ramsar Convention?

- **Definition of Wetlands** : Includes **natural or artificial, permanent or temporary marshes, peatlands and water bodies, with fresh, brackish or saline water,** including marine areas up to six metres deep at low tide.
- **Three Pillars of the Convention:**
  - **Wise Use:** Maintaining the ecological character of all wetlands through sustainable ecosystem-based management.
  - **Ramsar Site Designation:** Identifying and effectively managing Wetlands of International Importance.
  - **International Cooperation:** Promoting cooperation on shared wetlands, transboundary water systems and migratory species.
- **Designation Process in India:** The State Government or UT Administration identifies eligible wetlands, while the **Ministry of Environment, Forest and Climate Change (MoEFCC)** coordinates approval and submits proposals to the Ramsar Secretariat.



Ramsar Criteria for Identifying Wetlands of International Importance

- **Group A – Wetland types**
  - **Criterion 1:** A wetland is internationally important if it contains a **representative, rare, or unique example** of a natural or near-natural wetland type found within the appropriate biogeographic region.
- **Group B – Biological diversity: Species and ecological communities**
  - **Criterion 2:** It supports **vulnerable, endangered, or critically endangered species** or threatened ecological communities.
  - **Criterion 3:** It supports populations of **plant and/or animal species important for maintaining biological diversity** of a particular biogeographic region.
  - **Criterion 4:** It supports plant and/or animal species at a **critical stage in their life cycles** or provides refuge during adverse conditions.
- **Group B – Biological diversity: Waterbirds**
  - **Criterion 5:** It regularly supports **20,000 or more waterbirds** .
  - **Criterion 6:** It regularly supports **1% of the individuals in a population of one species or subspecies of waterbird** .
- **Group B – Biological diversity: Fish**
  - **Criterion 7:** It supports a significant proportion of **indigenous fish subspecies, species or families** , life-history stages, species interactions and/or populations that represent wetland benefits or values and contribute to global biological diversity.
  - **Criterion 8:** It is an important **source of food for fishes, spawning ground, nursery and/or migration path** on which fish stocks within or outside the wetland depend.
- **Group B – Biological diversity: Other taxa**
  - **Criterion 9:** It regularly supports **1% of the individuals in a population of one species or subspecies of wetland-dependent non-avian animal species** .

Benefits of Designating Wetlands to Ramsar List

- **International Representation:** Provides countries a platform to raise concerns at the principal intergovernmental forum on wetland conservation and wise use. India introduced the resolution on **“Promoting Sustainable Lifestyles for the Wise Use of Wetlands”** at the Ramsar COP15, which

was adopted with broad support.

- **Enhanced Visibility and Recognition:** Increases the international visibility of Ramsar Sites and strengthens support for their conservation and wise use.
- **International Standards and Guidance:** Provides access to internationally accepted standards for identifying wetlands of international importance, along with guidance on wise use and management planning.
- **Expert Assistance:** Enables countries to seek expert advice on national and site-specific wetland conservation and management challenges through the Ramsar Secretariat, its collaborators and Ramsar Advisory Missions.
- **International Cooperation and Financial Support:** Promotes international cooperation in wetland conservation and management and can facilitate project support through Ramsar's small-grant programmes and multilateral and bilateral funding agencies.

Commitments After Designation

- **Wise Use and Conservation:** Designation commits the concerned State or Union Territory to the **wise use and conservation** of the wetland and its resources.
- **Legal and Management Framework:** The Ramsar Site is deemed notified under the **Wetlands (Conservation and Management) Rules, 2017** , with guidelines recommending formal notification along with an **Integrated Management Plan** .
- **Periodic Monitoring and Reporting:** The **Ramsar Information Sheet (RIS)** must be updated every six years, while the ecological character of the site must be monitored regularly.
- **National Wetland Network:** These criteria and commitments have supported the development of India's large and diverse network of **Ramsar Sites** .

Montreux Record

- **Meaning:** The **Montreux Record** is a register of Ramsar Sites where changes in ecological character have occurred, are occurring, or are likely to occur.
- **Purpose:** It identifies wetlands requiring **priority conservation attention** and supports monitoring, remedial action and resource allocation.
- **Inclusion and Removal:** Sites are included in consultation with the concerned country and relevant scientific experts and may be removed once **successful remedial measures** are documented or the threat of ecological change no longer exists.

What is the Ecological and Socioeconomic Significance of Ramsar Sites?

- **Biodiversity Conservation** : Wetlands provide **breeding, feeding and migratory habitats for numerous species** , including threatened waterbirds, fish and aquatic mammals.
  - **Example: Chilika Lake (Odisha)** supports approximately 118 fish species and important migratory bird habitats.
- **Climate Resilience:** Wetlands, particularly mangrove ecosystems, act as natural buffers against floods,

cyclones, storm surges and coastal erosion.

  - **Example: Sundarban Wetland** protects coastal settlements from extreme weather events and saltwater intrusion.

- **Water Security:** Wetlands contribute to groundwater recharge, freshwater storage, water purification and hydrological regulation.
- **Sustainable Livelihoods:** Wetlands support fisheries, agriculture, tourism and traditional occupations, improving local economic security.
  - **Example: East Kolkata Wetlands** naturally treat wastewater through approximately 4,000 hectares of fish ponds, supporting fisheries and local livelihoods.
- **Species Recovery:** Protection of wetland ecosystems supports the conservation and recovery of threatened species.
  - **Example: Beas Conservation Reserve (Punjab)** supports India's only known population of the Indus River dolphin.
- **Community Participation:** Local stewardship strengthens long-term conservation through traditional ecological knowledge and collective action.
  - **Example: Khichan Wetland (Rajasthan)** supports over 22,000 wintering demoiselle cranes, with active protection by local communities.
- **International Cooperation:** Ramsar designation promotes access to scientific expertise, international conservation standards and opportunities for collaborative wetland management.

Conclusion

India's expanding Ramsar network reflects growing recognition of wetlands as critical natural infrastructure for biodiversity conservation, water security and climate resilience. Strengthening scientific monitoring, community participation and integrated ecosystem management is essential to translate international recognition into lasting ecological conservation outcomes.

Drishti Mains Practice Question

Wetlands are critical ecological assets that contribute to biodiversity conservation, climate resilience and livelihood security. Examine.

Frequently Asked Questions (FAQs)

Q1. How many Ramsar Sites are there in India as of October 2026?
India has 103 Ramsar Sites, covering more than 13.87 lakh hectares.

Q2. Which state has the highest number of Ramsar Sites in India?
Tamil Nadu has the highest number with 20 sites, followed by Uttar Pradesh with 14 sites.

Q3. What are the latest Ramsar Sites added in India?
Sukhna Wetland in Chandigarh and Sauj Jheel in Uttar Pradesh were added in October 2026.

Q4. What is the Montreux Record?
It is a register of Ramsar Sites where changes in ecological character have occurred, are occurring or are likely to occur, requiring priority conservation attention.

Q5. Which are India's first and largest Ramsar Sites?
Chilika Lake and Keoladeo National Park were India's first Ramsar Sites, while Sundarban Wetland in West Bengal is the largest.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/cEEFaVouSqI]

UPSC Civil Services Examination Previous Year Questions (PYQs)

Prelims

Q. Consider the following statements: (2019)

1. Under Ramsar Convention, it is mandatory on the part of the Government of India to protect and conserve all the wetlands in the territory of India.
2. The Wetlands (Conservation and Management) Rules, 2010 were framed by the Government of India based on the recommendation of Ramsar Convention.
3. The Wetlands (Conservation and Management) Rules, 2010 also encompass the drainage area or catchment regions of the wetlands as determined by the authority.

Which of the statements given above is/are correct?

a) 1 and 2 only

b) 2 and 3 only

c) 3 only

d) 1, 2 and 3

Ans: (c)

Q. If a wetland of international importance is brought under the 'Montreux Record', what does it imply? (2014)

(a) Changes in ecological character have occurred, are occurring or are likely to occur in the wetland as a result of human interference.

(b) The country in which the wetland is located should enact a law to prohibit any human activity within five kilometres from the edge of the wetland.

(c) The survival of the wetland depends on the cultural practices and traditions of certain communities living in its vicinity and therefore the cultural diversity therein should not be destroyed.

(d) It is given the status of 'World Heritage Site.'

Ans: (a)

Mains

Q. What is wetland? Explain the Ramsar concept of 'wise use' in the context of wetland conservation. Cite two examples of Ramsar sites from India. **(2018)**

Q. How does biodiversity vary in India? How is the Biological Diversity Act, 2002 helpful in conservation of flora and fauna? **(2018)**

National SARAS Mela 2026

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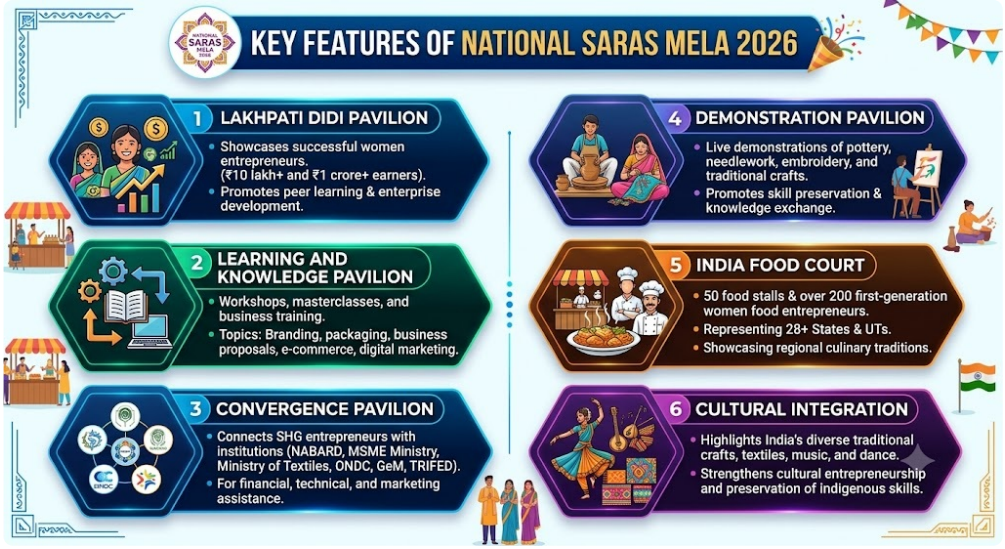
Source: PIB

Why in News?

The **National SARAS Mela 2026, organised in Vijayawada, Andhra Pradesh** , marks the **first National SARAS Mela outside Delhi-NCR** , bringing together women entrepreneurs from 34 States and Union Territories to promote **Self-Help Group (SHG)** enterprises, market access and sustainable rural livelihoods under **DAY-NRLM** .

What are the Key Facts About SARAS Mela?

- **About:** SARAS Mela is a **government-supported platform** organised under the **Deendayal Antyodaya Yojana–National Rural Livelihoods Mission (DAY-NRLM)** to connect rural women entrepreneurs particularly Self-Help Groups (SHGs), with consumers, institutional buyers and wider markets.
- **Evolution:** The first **National SARAS Mela was organised in Delhi in 1999, with approximately 65 beneficiaries** . It has since expanded to national, regional, state and district levels.
- **Major Achievements:** A total of 117 National SARAS Melas have been organised, with annual sales exceeding ₹200 crore.
  - States and UTs are expected to organise at least **two State-level SARAS Melas annually,** contributing to an average of around **50 events nationwide.**
- **Products Showcased: Handlooms, handicrafts, traditional textiles, natural products, processed foods and regional cuisines** , alongside an India Food Court, cultural showcases and dedicated learning spaces.
- **Institutional Support:** The initiative involves collaboration with institutions such as **NABARD , the National Institute of Rural Development and Panchayati Raj (NIRDPR)** and other government and private-sector stakeholders.
- **Digital Market Access:**
  - **eSARAS Platform :** An e-commerce initiative enabling customers to discover and purchase products made by SHG entrepreneurs beyond physical exhibitions.
  - **SARAS Mela Digital App (2025) :** Facilitates SHG nominations, attendance management and event-related information.
  - **Digital Payments:** Promotes UPI-based transactions, while retaining alternative payment arrangements for financial inclusivity.



What is the Role of DAY-NRLM and Lakhpati Didi in Rural Women Empowerment?

- **DAY-NRLM:** A **flagship rural poverty alleviation programme that promotes women's economic empowerment through SHGs** , providing access to credit, skills, technology, institutional support and sustainable livelihood opportunities.
  - SARAS Melas supports DAY-NRLM by strengthening product branding, direct sales, buyer–seller linkages and access to national markets.
- **Lakhpati Didi Initiative:** An **outcome-oriented initiative under DAY-NRLM** aimed at enabling SHG women to achieve sustainable annual household **incomes of at least ₹1 lakh** .
  - Income sustainability is assessed over at least four agricultural seasons and/or business cycles.
  - The target increased from 2 crore in 2023 to 3 crore in 2024, with the 3-crore milestone achieved by December 2025.
  - In **February 2026,** the government announced an enhanced target of 6 crore Lakhpati Didis.
- **Enterprise Development:** SARAS Melas support the transition from subsistence livelihoods towards women-led microenterprises, enabling income diversification, business expansion and sustainable market participation.

Conclusion

SARAS Mela 2026 reflects a shift from traditional livelihood promotion towards market-oriented rural entrepreneurship and women's economic empowerment. By integrating SHG enterprises, institutional support, digital marketplaces and the Lakhpati Didi initiative, it strengthens inclusive rural development and advances the vision of **Atmanirbhar Bharat** .

Drishti Mains Question

Self-Help Groups have emerged as important instruments of women's economic empowerment and rural livelihood diversification in India. Discuss.

Frequently Asked Questions (FAQs)

Q1. What is SARAS Mela?
SARAS Mela is a government-supported exhibition and marketing platform under DAY-NRLM, promoting products manufactured by rural Self-Help Group entrepreneurs.

Q2. Where was the first National SARAS Mela outside Delhi-NCR organised?
The first National SARAS Mela outside Delhi-NCR was organised in Vijayawada, Andhra Pradesh, in October 2026.

Q3. What is the Lakhpati Didi Initiative?
It is an initiative under DAY-NRLM aimed at enabling women SHG members to achieve sustainable annual household incomes of at least ₹1 lakh.

Q4. What is the eSARAS Platform?
eSARAS is an online marketplace that connects products made by rural SHG entrepreneurs with consumers across India.

Q5. What is the revised target under the Lakhpati Didi Initiative?
The government announced a revised target of 6 crore Lakhpati Didis in February 2026.

[Watch Video on YouTube: ▶ https://www.youtube.com/embed/KsPjaU9n4K0]

UPSC Civil Services Examination Previous Year Questions (PYQs)

Prelims

Q. How does the National Rural Livelihood Mission seek to improve livelihood options of rural poor? (2012)

1. By setting up a large number of new manufacturing industries and agribusiness centres in rural areas
2. By strengthening ‘self-help groups’ and providing skill development
3. By supplying seeds, fertilizers, diesel pump-sets and micro-irrigation equipment free of cost to farmers

Select the correct answer using the codes given below:

(a) 1 and 2 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3

Ans: (b)

Mains

Q. Can the vicious cycle of gender inequality, poverty and malnutrition be broken through microfinancing of women SHGs? Explain with examples. **(2021)**

PHP Vulnerabilities: Cybersecurity Threats in the AI Era

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Source: BBC

Why in News?

Two critical vulnerabilities in PHP (Hypertext Preprocessor), collectively known as **PHP2Shell,** have raised global cybersecurity concerns due to their potential to enable remote code execution and compromise sensitive data on affected servers.

What is the PHP Vulnerability?

- **About:** PHP (Hypertext Preprocessor) is an **open-source, back-end programming language** that runs on web servers, originally developed in the early 1990s.
- **Scale:** It is the foundational backbone of the internet, powering **70% of the world's websites** , including Wikipedia, WordPress, online shops, and popular frameworks like Laravel and Drupal.
- **Maintenance:** PHP is developed and maintained by its open-source community, with support from the PHP Foundation.
- **PHP Vulnerabilities:** The vulnerabilities being tracked as CVE-2026-109618 and CVE-2026-109619, collectively known as PHP2Shell, have a reported severity score of 9.2/10.
  - **CVE-2026-109618:** This vulnerability involves improper handling of **out-of-memory errors** , which can corrupt memory during request processing.
  - **CVE-2026-109619:** This vulnerability involves accessing **previously freed memory** , potentially allowing attackers to enable **unauthenticated** **Remote Code Execution (RCE)** .
  - These vulnerabilities potentially allow attackers to take control of web servers and compromise the **Confidentiality, Integrity and Availability (CIA) Triad.** Hackers have actively tried to use these flaws to install variants of the **Mirai botnet** .
  - The vulnerabilities were identified using AI-powered code analysis, highlighting **how AI can strengthen cybersecurity by detecting flaws** while also being misused to develop cyberattacks.
- **User Awareness:** While backend developers bear the primary responsibility for patching, end-users are advised to remain vigilant against **phishing attacks** that exploit the public panic surrounding massive cyber incidents.
- **Impact on Critical Information Infrastructure (CII):** The disclosure prompted precautionary shutdowns and emergency maintenance of websites and digital services worldwide.
  - Major entities like Wikipedia, the UK and Finnish governments, and telecom giants suspended services.

- Notably, Japan's SoftBank had to **temporarily suspend emergency mobile alerts for natural disasters (earthquakes/tsunamis)** to protect its systems, highlighting how software bugs can cripple real-world emergency response systems.
- In countries like India, **CII** heavily relies on older systems built for functionality rather than security. These **legacy systems are exceptionally vulnerable to AI-enabled exploit generation.**

Click here to Read: AI Standards Body and Agentic Misalignment Risks

Frequently Asked Questions (FAQs)

1. What are PHP vulnerabilities?
They are security flaws in PHP software that may allow attackers to compromise web servers. The vulnerabilities described in the article reportedly enable Remote Code Execution (RCE).

2. What is the CIA Triad in cybersecurity?
It comprises Confidentiality, Integrity and Availability, the three core principles of information security.

3. What is CERT-In?
The Indian Computer Emergency Response Team is India's national agency for responding to cybersecurity incidents and issuing security advisories.

4. What is Critical Information Infrastructure (CII)?
CII refers to computer resources whose incapacitation or destruction could have a debilitating impact on national security, the economy, public health or safety.

5. What is a Mirai botnet?
Mirai is malware that hijacks vulnerable Internet of Things (IoT) devices to create botnets capable of launching large-scale DDoS attacks.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/xwdY7Plit70**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. The terms 'WannaCry, Petya and EternalBlue' sometimes mentioned in the news recently are related to (2018)

(a) Exoplanets
(b) Cryptocurrency
(c) Cyber attacks
(d) Mini satellites

Ans: (c)

Q. In India, it is legally mandatory for which of the following to report on cyber security incidents? (2017)

1. Service providers
2. Data centres
3. Body corporate

Select the correct answer using the code given below:

(a) 1 only
(b) 1 and 2 only
(c) 3 only
(d) 1, 2 and 3

Ans: (d)

Mains:

Q. What are the different elements of cyber security? Keeping in view the challenges in cyber security, examine the extent to which India has successfully developed a comprehensive National Cyber Security Strategy. **(2022)**

Q. Discuss the potential threats of Cyber-attack and the security framework to prevent it. **(2016)**

Ashoke Sen Wins 2026 Oskar Klein Medal

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Source: TH

Renowned Indian theoretical physicist **Ashoke Sen became the first Indian to receive the Oskar Klein Medal (2026)** for his pioneering contributions to string theory, quantum gravity and

black hole physics.

- **About Oskar Klein Medal:** Established in **1988** to honour Swedish theoretical physicist **Oskar Klein (1894–1977)** , it is awarded by **Stockholm University and the Nobel Committee for Physics** to recognise outstanding contributions to theoretical physics.
- **Oskar Klein's Contribution:** Building on Theodor Kaluza's work, Oskar Klein advanced Kaluza-Klein theory. This theory sought to unify gravity and electromagnetism through an **additional spatial dimension** , influencing later developments in string theory.
- **About Ashoke Sen:** He is the **ICTS-Infosys Madhava Chair Professor** at the International Centre for Theoretical Sciences (ICTS), Bengaluru. He delivered the **2026 Oskar Klein Memorial Lecture** in Stockholm on **“Gravity from Strings and Strings from Gravity.”**
- **Key Contributions:**
  - **S-Duality:** Helped establish that apparently **different descriptions of physical interactions can be mathematically equivalent** , particularly strongly and weakly coupled theories.
  - **Sen Conjectures (1999):** Proposed **three conjectures on tachyon condensation and unstable D-branes** , advancing the understanding of string dynamics.
  - **String Field Theory:** Advanced the mathematical understanding of **string interactions and dynamics** .
  - **Black Hole Physics:** Developed methods to explain **black hole entropy through the counting of microscopic quantum states** .
- **Major Honours:** ICTP Prize (1989), **Shanti Swarup Bhatnagar Prize** (1994), Padma Shri (2001), **Infosys Prize** (2009), Breakthrough Prize in Fundamental Physics (2012), Padma Bhushan (2013) and Dirac Medal (2014).

About String Theory

- **Concept:** A theoretical framework proposing that **fundamental particles are tiny vibrating strings rather than point-like particles** , with different vibrations representing different particles.
- **Objective and Significance:** Seeks to **reconcile quantum mechanics and general relativity** , providing a unified framework for understanding **gravity, other fundamental forces and black hole physics** .
- **Limitation:** Despite its mathematical sophistication, **string theory remains experimentally unverified** .



Read More: Nobel Prize in Physics 2026: IceCube Neutrino Observatory

Madras HC Condemns use of Children in Political Campaigns

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Source: TH

The Madras High Court observed that Parliament should consider **whether involving children in political campaigning should be a ground for setting aside a candidate's election,** raising concerns about child rights, voter autonomy and electoral ethics.

- **Legal Standing:** The Madras High Court clarified that an appeal to children to persuade their parents does not amount to **"undue influence" or a "corrupt practice" under Section 123(2) of the Representation of the People Act (RPA), 1951** .
  - The court determined that **persuasion by children does not overpower the free will or volition of adult voters** , who retain the ultimate agency to make informed electoral choices.
- **Judicial Criticism:** The court strongly condemned the practice, calling it **"in bad taste," reflective of "poor parenting," and a "cheap and politically motivated act"** .
- **Election Commission of India (ECI) Directives:** The ECI maintains a strict **"zero tolerance"** policy regarding the use of children in any election-related activities, including rallies, carrying party flags, or slogan shouting.
  - However, merely accompanying a parent or guardian near a political leader, without participating in campaigning, does not constitute a violation.
  - Political parties and candidates must comply with the **Child and Adolescent Labour (Prohibition and Regulation) Act, 1986,** as amended in 2016. District Election Officers and Returning Officers are responsible for ensuring compliance by election machinery.
- **The Legislative Gap:** While ECI guidelines exist as administrative advisories, the **Representation of the People Act, 1951** , lacks explicit provisions that automatically void a candidate's election for violating these specific child-protection directives under **Section 100** (grounds for declaring an election void).
- **International Framework** : The **UN Convention on the Rights of the Child (UNCRC)** recognises children's rights to expression and participation, while **UNICEF's child-participation framework distinguishes** meaningful civic participation from manipulation and tokenism.

Read more: Engaging India's Youth in Voting

Upgraded Agri Welfare Portal Launched for Farmers

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Source: PIB

The **Ministry of Agriculture and Farmers Welfare launched the upgraded Agri Welfare Portal** to provide farmers with single-window, mobile-based access to government schemes, financial assistance and agricultural information.

- **About:** It is a **single-window digital platform** providing farmers access to **government schemes, financial assistance and reliable agricultural information** through mobile phones.
- **Single-Window Access:** The portal brings information and services related to **schemes, subsidies, rules and digital services** onto one platform, reducing the need to visit multiple offices or websites.
- **Key Schemes:** It provides access to major initiatives including **PM-KISAN** , **AgriStack** **and** **Pradhan Mantri Fasal Bima Yojana (PMFBY)** through a common interface.
- **Transparency and DBT: Online eligibility checks, direct applications and Direct Benefit Transfer (DBT)** aim to simplify processes, improve transparency and reduce intermediaries.
- **Farmer-Centric Design:** The portal is **lightweight, mobile-friendly and accessible on basic smartphones** , even in areas with weak internet connectivity, providing information on **weather, scientific farming practices, seeds, fertilisers and market prices** .
- **Agricultural Ecosystem:** It also provides a common digital platform connecting **Farmer Producer Organisations, Primary Agricultural Credit Societies, scientists, researchers and startups** .
- **Capacity Building:** **Krishi Vigyan Kendras** will conduct training campaigns to help farmers, particularly small and marginal farmers, effectively use the portal and other digital tools.
- **Digital Standards:** The portal has been developed in accordance with the **Digital Brand Identity Manual 3.0** and **Guidelines for Indian Government Websites 3.0** , focusing on security, accessibility and usability.

Read More: **Kisan Sarathi Platform**